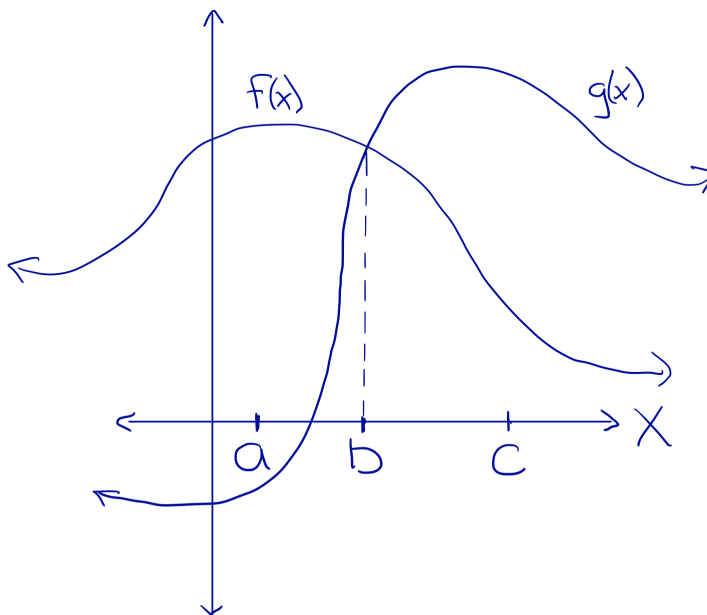


Goal. Use the integral to find the area of various regions. In each problem, be sure to draw a graph of the functions.

1. A sketch of the functions $f(x)$ and $g(x)$ is drawn below:



Here, $\int_a^b |f(x) - g(x)| dx = 8$ and $\int_b^c |f(x) - g(x)| dx = 12$.

(a) What is $\int_a^c (f(x) - g(x)) dx$?

(b) What is $\int_a^c |f(x) - g(x)| dx$?

2. **True or false:** $\int_0^1 f(x) \cdot g(x) dx = \int_0^1 f(x) dx \cdot \int_0^1 g(x) dx$. Either justify why this statement is true, or give an example showing it is not true.
3. What equation describes a circle centered at $(x, y) = (2, 4)$ with radius 5?
4. Sketch a region whose area is represented by $\int_{-\sqrt{2}/2}^{\sqrt{2}/2} (\sqrt{1-x^2} - |x|) dx$ and evaluate using geometry.
5. Write your own problem that can be interpreted as the *area between two curves* where the final answer is 4π . Then, show your problem answer is 4π .